Physical qualification: what can be identified?

16 October 2026 — MP6-Q1; no traffic experiment or parameter selection

Status. The selected conditional model is unchanged. Its forward domain is qualified at WF-Q1. This note separates its physical meanings and gives exact, conditional observation identities. It does not establish an empirical parameter range, noisy-data identifiability or applied MP6.

1. A boundary recovery quantity is not a vehicle trajectory

Retain $K = wk_j$, $c(s) = Ks/(w+s)$, $U(f) = wf/(K-f)$ and $0 < R \le C(v_-) = c(v_-) < K$. In the selected node,

$$c_E = 1 + \delta + \chi g_A[u_A - u_E]_+, \quad c_A = 1, \quad B = R - p, \qquad g_A + c_E g_E \le B. \tag{1}$$

The previously proved positive-budget domain makes clipping inactive. Free marks are $u_j = v$; queued marks are $u_j = U(g_j)$. Their use as actual entry/passage speed is an equilibrium approximation, not a consequence of conservation. χ is fixed in physical units across advice.

For an isolated entry with mark $u \ge 0$, define

$$\begin{aligned} S(\theta;u) &= \sqrt{(w+u)^2 + 2wa_r\theta}, \quad V(\theta;u) = S(\theta;u) - w, \\ Q_{\text{iso}}(\theta;u) &= K\left[\theta - \frac{S(\theta;u) - w - u}{a_r}\right], \\ \partial_\theta Q_{\text{iso}} &= K(1 - w/S) = c(V), \quad D(\theta;u) = [R - c(V)]_+. \end{aligned} \tag{2}$$

The derivative identity explains the retained profile. V parametrizes the boundary passing rate; it is not the admitted vehicle's speed. For comparison, a vehicle accelerating constantly before a cap has speed $u + a_r\theta$ along its own trajectory, which differs from V. Actual acceleration observations constrain the source interpretation of a_r; observed discharge tests (2). Fitting V directly to measured vehicle speeds would test a different assertion.

This distinction is explicit in the accompanying interpretation of equation (1) of Leclercq et al. (2016). Their congested insertion-speed assumption in Section 3.2 gives a restricted precedent for a flow-to-speed relation, not a proof of every transient stream mark used here.

Units: g_j, B, p, D, K are veh-eq/s; u_j, w, V are m/s; a_r is m/s^2; δ is dimensionless; χ is s^2/(veh-eq m). The reference mass $m = 1$ veh-eq is physical normalization, not an integer service packet.

2. What distinguishes recovery from current maneuver work?

With the selected recent-mass renewal, Y_E is cumulative admitted E and $T(\xi)$ its mass-rank admission time:

$$p(t) = \frac{1}{m} \int_{[Y_E(t)-m]_+}^{Y_E(t)} D(t - T(\xi); u_E(T(\xi)))\, d\xi. \tag{3}$$

If E has finished entering and A remains queued, $g_E = 0$ and its current maneuver work vanishes. Work conservation in the selected node gives

$$g_A = B = R - p, \qquad p = R - g_A. \tag{4}$$

This identifies the account's value on that ideal saturated tail if R is independently known. It does not identify every profile parameter from one tail, guarantee such a tail exists in the primary cohort, or make a microscopic individual headway a continuous instantaneous rate. The later execution proposal must specify observation scale and uncertainty. If A is free, low g_A alone does not identify B.

At a binding mixed-service observation with $g_E > 0$, set

$$X = g_A[u_A - u_E]_+, \qquad Z = \frac{R - p - g_A}{g_E} - 1. \quad \Longrightarrow \quad Z = \delta + \chi X. \tag{5}$$

After recovery and marks are constrained separately, two distinct covariate values have determinant $X_2 - X_1$ and give

$$\chi = \frac{Z_2 - Z_1}{X_2 - X_1}, \qquad \delta = \frac{X_2 Z_1 - X_1 Z_2}{X_2 - X_1}. \tag{6}$$

Admissible estimates must still obey the selected nonnegative and monotone-root conditions. Equal X values do not separate the coefficients; all $X = 0$ gives no information about χ. With nonbinding service, the counterpart of (5) is only an inequality.

At a single observation without independent recovery information, the accounting ambiguity is explicit:

$$\delta' = \delta + \varepsilon, \qquad p' = p - \varepsilon g_E \quad \Longrightarrow \quad p' + g_A + (1 + \delta' + \chi X) g_E = p + g_A + (1 + \delta + \chi X) g_E. \tag{7}$$

This is not a second admissible O-R trajectory or a global non-identifiability theorem: the primed account need not satisfy (3). It shows why discharge accounting alone does not establish the physical partition. Distinct tails, mixtures and joint predictions are needed.

The profile already removes service through (3). Adding its integrated loss again to c_E would charge that same modeled loss twice. Whether fitted immediate work represents a distinct real burden still requires evidence; algebra alone cannot establish that physical fact.

3. Which preparation and longitudinal scales are visible?

For one constant advice, use traveled preparation distance z, remaining ending density $e(z)$ and the selected H1 relation

$$\frac{de}{dz} = -\frac{1}{h}\min\left\{e, \frac{\Delta + e}{\gamma}\right\}, \qquad \Delta = \frac{K}{w+v} - r. \tag{8}$$

h is an opportunity length and γ is dimensionless receiving penalty. On an informative donor-limited interval,

$$e(z) = e(z_0)e^{-(z-z_0)/h}, \qquad -\frac{d\log e}{dz} = \frac{1}{h}. \tag{9}$$

On a receiving-limited interval with fixed v, r,

$$e(z) + \Delta = [e(z_0) + \Delta]e^{-(z-z_0)/(\gamma h)}, \qquad -\frac{d\log(e+\Delta)}{dz} = \frac{1}{\gamma h}. \tag{10}$$

Thus that branch identifies a product. The transformation $h' = sh$, $\gamma' = \gamma/s$ preserves (10), as long as the branch and admissibility remain unchanged. Donor-limited observations can constrain h first. If no receiving-limited behavior is observed, the penalty may only be bounded. A single final E count generally cannot identify the whole law. These are ideal branch-conditional statements, not statistical estimators for arbitrary noisy spatial observations.

For the earlier stationary correspondence, let d be vehicle length plus standing gap and τ_d a desired time gap. In the documented ideal constant-speed Krauss case,

$$k_{\text{ideal}}(s) = \frac{1}{d + \tau_d s}, \qquad f_{\text{ideal}}(s) = \frac{s}{d + \tau_d s}. \tag{11}$$

Choosing $k_j = 1/d$ and $w = d/\tau_d$ gives

$$\frac{K}{w+s} = \frac{1}{d+\tau_d s}, \qquad c(s) = \frac{s}{d+\tau_d s}. \tag{12}$$

This is a stationary curve identity at actual speed s. It does not prove that native transient waves propagate at this w, that advice is instantly realized, or that a separately served lane stream has speed $U(g_j)$. SUMO's documented headway variations and insertion effects matter to any later admission. No runtime capacity was measured in this review.

4. Recovery scale and joint-domain implications

For $u < u_R = U(R)$ the isolated active-profile duration and integrated loss are

$$\Theta(u) = \frac{(w+u_R)^2 - (w+u)^2}{2wa_r}, \qquad W(u) = \int_0^{\Theta(u)} D(\theta; u)\, d\theta = \frac{K}{w+u_R}\frac{(u_R-u)^2}{2a_r}. \tag{13}$$

For $u \geq u_R$, both are zero. While the profile is active,

$$\frac{\partial D}{\partial \theta} = -\frac{Kw^2 a_r}{S^3}, \quad \frac{\partial D}{\partial u} = -\frac{Kw(w+u)}{S^3}, \quad \frac{\partial D}{\partial a_r} = -\frac{Kw^2\theta}{S^3}. \tag{14}$$

At fixed admission times and marks, stronger acceleration reduces each surviving profile and the account in (3). In the coupled traffic model, admissions and marks also change. This is not a theorem that complete TST or optimal advice varies monotonically with acceleration.

W has units veh-eq and Θ has units s. They are profile descriptors, not new current E charges. They constrain different aspects of recovery: a total loss alone confounds low insertion speed with slow acceleration; independent marks and time-resolved tails are more informative. O-R's retirement may truncate an older profile before acceleration removes it, so the isolated area is not the integrated account of an arbitrary history.

Joint admission. Require one common parameter set that explains preparation, actual speed marks, mixed discharge and recovery, and lies in the existing full-advice forward domain. In particular retain

$$v_+ r < R \leq C(v_-), \qquad M < k_j b, \qquad c_0 - \beta(1 + v_+/w) > 0, \qquad a_r > 0. \tag{15}$$

Other H1/T1 and far-lane assumptions remain as previously declared. $\chi = \beta/[C_\star v_\star]$ is anchored once, never refitted by advice or search box. m remains the selected physical normalization. (15) is mathematical admission, not an observed physical range. A good objective fit cannot replace the mechanism checks. No numerical value, tolerance, positive policy result or theory-only primary is selected here.

Queue-scope clarification. The earlier E non-requeue conclusion uses the monotone H1 pulse, free-speed/recovery rules and retained domain. Finiteness alone is insufficient: a bursty later E wave can requeue in a different model or a native finite cohort. Such an event cannot by itself be used to reject the independent reference.

Source ownership. Carey Section 5 motivates receiving work; Leclercq Sections 2 and 3.2 constrain the recovery and speed interpretations; official SUMO road-capacity documentation supports only the stationary correspondence. Equations (4)–(10), the admission order and joined-law qualification are our deductions. No source validates the complete node. See the accompanying viewed-reference list and physical admission contract. Applied MP6 and Step 3 remain open; Step 4 is unopened.